Making a Piecewise Model Continuous

Answer Key

AP Calculus AB/BC

Quick Answers

1. 40 degrees
2. 5
3. 10
4. 6 m
5. $\frac{1}{6}$
6. a = 4, b = −4
7. (a) the value of a = 50, —
8. a = 8, b = 2

What is verified

10 of 11 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problem 7. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC
LIM-2 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 11 tagged checks.

1. Oven: $T(t) = 20t + a$ for $0 \le t < 5$ and $T(t) = 150 - 2t$ for $t \ge 5$, in degrees Celsius. Find a so T is continuous at $t = 5$.

Solution: Continuity at $t = 5$ needs the two one-sided limits to agree.

$$\lim_{t \to 5^-} T(t) = 20(5) + a = 100 + a$$
$$\lim_{t \to 5^+} T(t) = 150 - 2(5) = 140$$

Set them equal: $100 + a = 140$, so $a = 40$. (Check: the second piece already gives $T(5) = 140$, so all three conditions hold.) $\boxed{a = 40 \text{ degrees}}$

2. Tank: $V(x) = 4x + 6$ for $0 \le x < 6$ and $V(x) = kx$ for $x \ge 6$, in litres. Find k so V is continuous at $x = 6$.

Solution: The left-hand limit at $x = 6$ is $4(6) + 6 = 30$, and the right-hand limit is $k(6) = 6k$. Continuity forces

$$30 = 6k \quad \Longrightarrow \quad k = 5.$$

So the tank holds 30 litres at the six-minute mark and the second stage must start from that same 30 litres. $\boxed{k = 5}$

3. Sensor: $r(x) = \dfrac{x^2 - 25}{x - 5}$ for $x \neq 5$, with $r(5) = c$. Choose c so r is continuous at $x = 5$.

Solution: A removable discontinuity is filled by the limit, so compute it. Direct substitution gives $\frac{0}{0}$; factor and cancel:

$$\frac{x^2 - 25}{x - 5} = \frac{(x-5)(x+5)}{x - 5} = x + 5 \qquad (x \neq 5).$$

Hence $\lim_{x \to 5} r(x) = 5 + 5 = 10$, and continuity requires c to equal that limit. $\boxed{c = 10}$

4. Drone: $h(t) = t^2 + a$ for $0 \leq t < 3$ and $h(t) = 4t + 3$ for $t \geq 3$, in metres. Find a so h is continuous at $t = 3$.

Solution: Match the one-sided limits at $t = 3$:

$$3^2 + a = 4(3) + 3 \qquad \Longrightarrow \qquad 9 + a = 15.$$

So $a = 6$. The drone is 15 metres up when the flight law changes, and a is the launch-pad offset that makes the first piece reach exactly that height. $\boxed{a = 6 \text{ m}}$

5. Calibration: $E(x) = \dfrac{\sqrt{x+7} - 3}{x - 2}$ for $x \neq 2$, with $E(2) = c$. Choose c so E is continuous at $x = 2$.

Solution: Again c must be the limit. Substitution gives $\frac{3-3}{0} = \frac{0}{0}$ and nothing factors, so multiply by the conjugate:

$$\begin{aligned} \frac{\sqrt{x+7} - 3}{x - 2} \cdot \frac{\sqrt{x+7} + 3}{\sqrt{x+7} + 3} &= \frac{(x+7) - 9}{(x-2)\left(\sqrt{x+7} + 3\right)} \\ &= \frac{x - 2}{(x-2)\left(\sqrt{x+7} + 3\right)} = \frac{1}{\sqrt{x+7} + 3} \qquad (x \neq 2). \end{aligned}$$

At $x = 2$ this is $\frac{1}{3+3}$, so the stored value must be $\boxed{c = \frac{1}{6}}$

6. Conveyor: $f(x) = x^2$ for $x < 2$, $f(x) = ax + b$ for $2 \leq x < 5$, $f(x) = 3x + 1$ for $x \geq 5$. Find a and b.

Solution: Each junction gives one equation.

$$\begin{aligned} \text{at } x = 2: \quad 2^2 &= a(2) + b & \Longrightarrow \quad & 2a + b = 4 \\ \text{at } x = 5: \quad a(5) + b &= 3(5) + 1 & \Longrightarrow \quad & 5a + b = 16 \end{aligned}$$

From the first equation $b = 4 - 2a$. Substituting into the second gives $5a + (4 - 2a) = 16$, so $3a = 12$ and $\boxed{a = 4}$

Back-substituting, $2(4) + b = 4$, so $\boxed{b = -4}$

Check: the middle piece is $4x - 4$, which equals 4 at $x = 2$ and 16 at $x = 5$ — both junctions close.

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7. Phone plan: $C(g) = 30 + 12g$ for $0 \leq g < 4$ and $C(g) = a + 7g$ for $g \geq 4$, in dollars. (a) Find a. (b) What would $a = 45$ do to a bill near 4 gigabytes?

Solution (a): The left-hand limit at $g = 4$ is $30 + 12(4) = 78$ and the right-hand limit is $a + 7(4) = a + 28$. Setting them equal,

$$78 = a + 28 \qquad \Longrightarrow \qquad a = 50.$$

So the second tier is $C(g) = 50 + 7g$, which is also 78 dollars at 4 gigabytes. $\boxed{a = 50}$

Solution (b) — instructor-judged, no machine check. With $a = 45$ the second piece gives $45 + 7(4) = 73$ dollars at 4 gigabytes while the first piece climbs to 78 dollars just below it: the bill *drops* by 5 dollars as usage crosses 4 gigabytes. Full credit identifies that 5-dollar downward jump and says what it means — a customer using slightly more than 4 gigabytes would be charged less than one using slightly less, which is why the plan is written to be continuous. Accept any wording that names the jump and its consequence; do not accept a bare "it would not be continuous" with no consequence stated.

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8. Challenge. $g(x) = \dfrac{x^2 - 16}{x - 4}$ for $x < 4$, $g(4) = a$, and $g(x) = bx$ for $x > 4$. Find a and b.

Solution: All three continuity conditions must hold at $x = 4$: the left-hand limit, the right-hand limit, and the value $g(4)$ must be one number.

Left-hand limit. Factor and cancel: $\dfrac{x^2 - 16}{x - 4} = \dfrac{(x-4)(x+4)}{x - 4} = x + 4$ for $x \neq 4$, so $\lim_{x\to 4^-} g(x) = 4 + 4 = 8$. Since $g(4) = a$ must equal that, $\boxed{a = 8}$

Right-hand limit. $\lim_{x\to 4^+} g(x) = b(4) = 4b$, and this must equal the same 8, so $4b = 8$ and $\boxed{b = 2}$

With $a = 8$ and $b = 2$ every piece meets at the single value 8, so g is continuous at $x = 4$.